

When Do Microphone-Array Geometry Priors Help Sound Event Localization and Detection? The Role of Temporal Inductive Bias

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Abstract

Hand-crafted microphone-array geometry priors are widely injected into deep sound event localization and detection (SELD) systems, often on the assumption that explicit spatial layout can only help. We test this assumption with a controlled one-bit intervention in the official DCASE 2024 SELD baseline: a geometry bias is toggled on or off inside an otherwise identical channel-attention block. Across two input modalities (MIC and FOA), three temporal backbones (CRNN+MHSA, Conformer, Transformer), and five paired seeds on real STARSS23 recordings, the prior’s effect on directional accuracy is not universal but graded by temporal inductive bias. It benefits the recurrent FOA model (-5.0° class-dependent localization error, $d_z = -1.03$), remains near-neutral to weakly helpful in the Conformer, and harms the pure-attention Transformer (up to $+5.7^\circ$, $d_z = +0.71$). A factorial test over the recurrent and Transformer extremes confirms an $\text{architecture} \times \text{prior}$ interaction ($F = 4.62$, $p = 0.039$; seed-matched paired contrast $p = 0.029$), and the recurrent FOA benefit replicates zero-shot on STARSS22 ($d_z = -1.50$, $p = 0.029$). A linear probe finds no significant change in linearly decodable

direction, indicating a behavioural rather than simple representational shift. Finally, a second injection mechanism that adds geometry as an early convolutional feature bias does not provide a clean cross-mechanism replication: its contrasts move across reruns, checkpoint-selection paths, and modalities, thereby bounding rather than strengthening the primary GCA claim. We conclude that explicit geometry priors are architecture-structured, injection-sensitive design choices rather than universal SELD improvements.

Keywords: sound source localization, direction of arrival, microphone array processing, ambisonics, deep learning, inductive bias

1. Introduction

Sound event localization and detection (SELD) is the array-acoustics problem of jointly estimating *what* sound events are active and *where* they originate—the latter a direction-of-arrival (DOA) estimation problem on a microphone array—and it underpins far-field automatic speech recognition, robot audition, augmented reality, and acoustic monitoring [1, 2]. SELD systems consume multichannel audio delivered either as a compact microphone array (MIC) or as first-order Ambisonics (FOA), and a recurring question in array signal processing is how much *array geometry* the network should be told explicitly. Many methods inject geometry through attention biases, positional encodings, or geometry-derived features [3], on the intuition that more spatial prior helps; a geometry-invariant line argues instead that inter-channel phase already suffices [4, 5].

This tension is rarely resolved in a controlled manner, because comparisons that motivate geometry priors typically change the architecture, the

input format, and the parameter count at once, confounding geometry with capacity and modality. We instead isolate the prior with a single *one-bit* intervention: inside a geometry-aware channel-attention (GCA) block we toggle one additive geometry bias on or off, holding the backbone, loss, optimizer, data, training recipe, and parameter count (to within 80 parameters) fixed. We embed this toggle in the official DCASE 2024 SELD baseline [6] and cross it with two input modalities (MIC, FOA) and *three* temporal backbones chosen to span the locality–globality spectrum: a convolutional-recurrent network with multi-head self-attention (CRNN with MHSA), a convolution–attention hybrid (Conformer), and a Transformer-only temporal stack.

Our central finding is that the prior’s effect on *direction* is not binary but *graded by the backbone’s temporal inductive bias*. As locality is progressively relaxed—from a recurrent stack, through a convolution–attention hybrid, to pure self-attention—the geometry effect moves from helpful, through near-neutral, to harmful. A factorial ANOVA over the recurrent and Transformer extremes confirms a significant architecture×prior interaction ($F = 4.62$, $p = 0.039$ at $n = 5$), while the Conformer consistently occupies the middle of the ordering rather than violating it. The most robust effect is the recurrent *benefit* on Ambisonics: the prior improves localization by 5.0° in-domain (a trend, $p = 0.082$) and replicates zero-shot on an independent dataset at full significance ($d_z = -1.50$, $p = 0.029$); the Transformer harm reproduces directionally in- and cross-domain ($+5.6$ – 5.7°). A second injection check, replacing the channel-attention bias with an early convolutional feature bias, sharpens the interpretation by limiting it: the Conformer remains near-neutral in some runs, whereas Transformer contrasts move across rerun and checkpoint-

selection paths, and the microphone-array Transformer can even reverse sign relative to GCA. The effect is therefore structured by architecture in the primary intervention, but not invariant to injection site or training-selection details. Orthogonally, the prior’s effect on *distance* follows a different rule: it improves relative distance error for Ambisonics across all three backbones but not the microphone array, so the architecture that governs direction is inert on distance—a dissociation (the modality×prior interaction a non-significant trend at $n = 5$) that underscores the prior’s context-dependent value. Because all effects are measured on real-world recorded acoustic scenes under matched compute, the resulting ordering gives array-processing practitioners an actionable rule for *when* to inject array geometry, not merely *whether* to.

1.1. Contributions

1. We show that the effect of an array-geometry prior on SELD *direction* is **graded by temporal inductive bias**: it helps a recurrent model, remains near-neutral to weakly helpful in a Conformer, and harms a Transformer, established by multi-seed paired ablation ($n = 5$) on STARSS23 across six modality×backbone cells, with a significant recurrent-versus-Transformer architecture×prior interaction ($p = 0.039$, seed-matched $p = 0.029$) and a monotone trend across all three backbones (Sections 4.2–4.3).
2. We establish a **dissociation**: the temporal architecture that governs the direction effect is inert on *distance*, whose prior benefit is independent of backbone and concentrated in Ambisonics by effect size (the modality×prior interaction being a non-significant trend at $n = 5$; Section 4.7).

3. We **anchor the pattern** with a zero-shot *cross-dataset replication* on STARSS22: the full architecture ordering holds out-of-domain, and the recurrent Ambisonic benefit is significant there ($d_z = -1.50$, $p = 0.029$) while the Transformer harm reproduces directionally (Section 4.4). A controlled linear probe finds *no* significant change in linearly decodable direction in any cell (Section 4.6), indicating the effect is behavioural rather than a change in linear feature content—a negative result we report in full.
4. We provide a **reproducible, statistically grounded protocol** (one-bit ablation, multi-seed paired tests, factorial interaction test, probing, and a second geometry-injection sensitivity check) and release the checkpoints needed to audit the claim, arguing that geometry priors should be ablated rather than assumed and that alternative injection sites should be treated as boundary conditions unless they are independently stabilized.

2. Related Work

2.1. Sound event localization and detection

SELD jointly answers *what* sound events occur, *when*, and *from where*. It was formalized as a convolutional-recurrent learning problem by Adavanne *et al.* [1] and has since been driven by the annual DCASE challenge and its evaluation protocol [2]. A central thread of progress concerns the *output parameterization*: the activity-coupled Cartesian DOA (ACCDOA) target folded detection and localization into a single regression [7], and its multi-source extension, Multi-ACCDOA with auxiliary duplicating permutation-invariant

training, resolved same-class polyphony [8]. The DCASE 2024 edition added *source distance estimation* to the task, and Krause *et al.* [6] released the official CRNN-with-self-attention baseline with a Multi-ACCDDOA head that we adopt here; building our study on this widely used reference keeps the geometry ablation interpretable and reproducible. On the architecture side, the field has moved from pure CRNNs toward attention-augmented and event-independent designs such as EINV2 [9], and the strongest recent challenge entries are large ResNet–Conformer ensembles [10]. These advances, however, typically bundle a new backbone, new features, and new training tricks together, which is precisely the confound our controlled design avoids.

2.2. Array geometry as a prior

How much array geometry a network should be told remains contested. *Geometry-aware* approaches inject explicit spatial information—array coordinates as mixed-data inputs [3], geometry-conditioned front-ends, or layout-derived attention biases—on the intuition that telling the model its microphone arrangement can only help. *Geometry-invariant* approaches argue the opposite: that inter-channel phase already carries the spatial cue, so explicit geometry is redundant or even limits cross-array generalization; representative work uses microphone positional encodings [4] or steered-response-power-style neural trackers that are trained to be array-agnostic [5]. These methods also differ in *where* geometry enters—from array coordinates concatenated to the input [3] to positional encodings that condition an array-agnostic front-end [4, 5]. Recent work has also moved physical spatial priors into the training objective, regularizing SELD outputs with theoretically or room-impulse-response-derived spatial covariance matrices [11]. These ap-

proaches make clear that geometry is not a single design choice but a family of constraints: it can enter as an input feature, a positional code, an attention bias, an early feature-map bias, or a loss-level structural prior. Our intervention is deliberately surgical within this space: a layout-derived term enters a single channel-attention block (Eq. (1)) and nothing else. The classical SELD features sit on this spectrum too: GCC-PHAT maps and Ambisonic intensity vectors encode geometry implicitly, so even `no_geom` is never fully geometry-blind. The decisive limitation of this literature is methodological: the two stances are almost always compared across *different architectures, feature sets, and datasets*, so the reported benefit (or harm) of geometry is confounded with capacity, input representation, injection site, and domain. We remove this confound for one primary geometry path by toggling a single geometry bias on and off inside one fixed architecture, and show that the answer is not universal but contingent on factors prior comparisons neither held fixed nor crossed.

2.3. Inductive bias and locality in temporal models

Our explanatory axis—how much *locality* a temporal backbone assumes—has a rich grounding in the inductive-bias literature [12]. Recurrent networks such as GRUs [13] process sequences with a strong locality/recency bias and limited capacity to recombine channels arbitrarily, whereas the Transformer [14] removes recurrence entirely and mixes all positions and channels globally through self-attention. The Conformer sits between them, reintroducing locality into an attention model via an interleaved depthwise-convolution module [15]. A growing body of vision and speech work shows that the *value* of an injected prior depends on how much structure the back-

bone already assumes: soft convolutional biases help data-hungry Transformers [16], and hybrid convolution–attention models trade off learned and built-in structure as a function of scale and data [17]. Relative-position formulations of attention make the same point from the opposite side, showing that adding locality to attention changes what it learns [18]. The unifying principle is *redundancy*: a prior helps when it supplies structure the backbone lacks and cannot cheaply learn (a low-capacity recurrent model under scarce real data), but is inert or harmful when the backbone already encodes it or can learn a better version (a globally-mixing Transformer). This predicts that a hand-crafted spatial prior should interact with the temporal backbone rather than help uniformly—exactly the graded effect we measure. To our knowledge we are the first to test this interaction directly in SELD, with a modality×backbone×prior factorial design.

2.4. Channel attention and the geometry bias

Attention has entered SELD both over time and over channels. Temporal self-attention on learned features is now standard [19]; over channels, squeeze/excitation recalibration [20] reweights feature maps. The geometry-aware channel-attention block we ablate augments channel attention with a layout-derived bias; removing *only* this bias while keeping the mechanism lets our comparison separate “channel attention with geometry” from “without geometry” and from “no channel attention,” isolating the prior from the mechanism that carries it.

2.5. Statistical rigor and positioning

Many SELD ablations—including the official baseline—report single runs with jackknife intervals, which cannot establish whether a small metric change is reliable rather than seed noise [21]. We instead adopt seed-matched paired tests, effect sizes, and a factorial interaction test as primary evidence, validate every effect zero-shot on an independent dataset [22], and stress-test on synthetic scenes [23], training in-domain on STARSS23 [24]. We do not chase the leaderboard; we ask whether a popular inductive bias delivers a consistent gain under matched budgets, treating the reproduced baseline (Section 4.1) as the reference.

3. Method

Our design philosophy is *intervention, not competition*: rather than proposing a new system, we hold a strong, widely used SELD backbone fixed. We then vary three binary factors in a fully crossed factorial design: the geometry prior (the variable of interest), the temporal architecture (how the network integrates the multichannel array signal over time), and the input modality. This lets any metric change be attributed to a named factor rather than to a tangle of co-varying design choices. This section describes the shared backbone (Sec. 3.1), the one-bit geometry intervention (Sec. 3.2), the temporal backbones (Sec. 3.3), and the metrics, statistics, and probing protocol (Sec. 3.4).

3.1. Shared backbone

We adopt the official DCASE 2024 Task 3 baseline [6] as the fixed scaffold. Audio sampled at 24 kHz is segmented into 5-second (250-frame) inputs.

The two input modalities differ only in their channel composition: the four-channel *microphone-array* (MIC) input is rendered as ten feature maps (4 log-mel spectrograms + 6 GCC-PHAT cross-correlation maps), while the *first-order Ambisonics* (FOA) input is rendered as seven maps (4 log-mel spectrograms + 3 acoustic intensity-vector maps). Both modalities thus carry their spatial information through standard, implicitly geometric features; the explicit prior we ablate is added on top.

A shared three-block 2-D convolutional stack (64 filters per block, batch normalization, ReLU) with frequency pooling factors $[4, 4, 2]$ and a single temporal pooling of 5 reduces each input to a 50-frame sequence of 128-dimensional embeddings. A temporal model (Sec. 3.3) then mixes this sequence over time before per-frame Multi-ACCDDOA heads [8] emit $3 \text{ tracks} \times 13 \text{ classes} \times (\text{activity, 3-axis Cartesian DOA, distance})$, trained with the auxiliary duplicating permutation-invariant (ADPIT) loss. All cells start from the official synthetic-data-pretrained weights and fine-tune for 60 epochs with the *identical* optimizer (Adam, learning rate 10^{-3} , batch size 32, dropout 0.05); we verify baseline parity against the published numbers in Section 4.1. Because the convolutional front-end, heads, loss, optimizer, data, and width are held fixed across every cell, capacity and recipe are controlled and the only differences are the three factors we vary.

3.2. The geometry prior as a one-bit intervention

The geometry-aware channel-attention (GCA) block performs single-head attention *over the microphone-channel dimension* of the input feature tensor, i.e. upstream of the shared convolutional front-end, letting the network reweight and recombine the array channels before convolution. Operating on

the raw array channels rather than post-convolution feature maps is what makes a microphone-pair geometry bias well-defined. Its geometry-aware variant injects a fixed, layout-derived feature $g_{ij} \in \mathbb{R}^4$ for each ordered channel pair (i, j) into the attention *keys*, so that geometry shapes the score before the dot product rather than as a constant added afterwards:

$$\text{att}(i, j) = \frac{q_i^\top (k_j + W_g g_{ij})}{\sqrt{d}} = \frac{q_i^\top k_j}{\sqrt{d}} + \frac{q_i^\top W_g g_{ij}}{\sqrt{d}}, \quad (1)$$

where $q_i, k_j \in \mathbb{R}^d$ are the learned query/key projections of the per-channel embeddings, d is their dimension, and $W_g \in \mathbb{R}^{d \times 4}$ (with bias) is the only learned geometry parameter. The pair feature g_{ij} is a deterministic, non-trainable function of the array layout (a registered buffer): for the microphone array it is $(\Delta x, \Delta y, \text{dist}, \text{bearing})$ between mics i and j in the array plane, and for Ambisonics it is the difference of the two channels' direction-of-maximum-response vectors together with a both-directional indicator, so the omnidirectional W channel and the $X/Y/Z$ first-order dipoles are correctly distinguished. Because the geometry enters only through the keys, its contribution to the logit, $q_i^\top W_g g_{ij} / \sqrt{d}$, is *content-dependent* (it is modulated by the query) rather than a fixed additive bias. The prior therefore tells the channel-attention layer which channels are spatially related *before* it has seen any data.

This block is the locus of our single intervention. The one-bit toggle is our sole design variable: **full** keeps the geometry projection W_g active, so Eq. (1) is geometry-aware; **no_geom** drops the $W_g g_{ij}$ term, reducing Eq. (1) to plain content-based channel attention. Crucially, the two variants differ by exactly the 80 parameters of W_g (a $\text{Linear}(4, 16)$ projection: 4×16 weights plus 16 biases) and are otherwise bit-for-bit identical in architecture, initialization,

data, and optimization, so a paired `full-no_geom` contrast isolates the *geometric prior* with essentially no change in capacity. To further separate the geometric bias from the attention *mechanism* that carries it, each cell also includes a **no-GCA** control that removes channel attention entirely, and the MIC+CRNN cell additionally includes a vanilla squeeze-and-excitation control [20]; both confirm that it is the geometry, not channel attention per se, that produces the effects we report. The toggle is well-defined and applied identically in both modalities, making the prior directly comparable across MIC and FOA.

3.3. Three temporal backbones spanning the locality spectrum

The hypothesis we test is that the geometry prior’s usefulness depends on how much *temporal locality* the backbone already assumes (Fig. 1). We therefore swap *only* the temporal model that sits between the shared convolutional front-end and the shared heads, choosing three backbones that span the locality–globality spectrum while being matched in width (128) and depth (≈ 4 temporal layers):

- **CRNN+MHSA** (the official baseline), the most *local* backbone: a two-layer bidirectional GRU [13] whose recurrence imposes a strong recency/locality bias and limits arbitrary channel recombination, followed by two layers of eight-head self-attention [19].
- **Conformer**, the *hybrid*: a linear projection followed by four Conformer blocks [15], each interleaving eight-head self-attention with a depthwise-convolution module (kernel 31) and feed-forward layers (di-

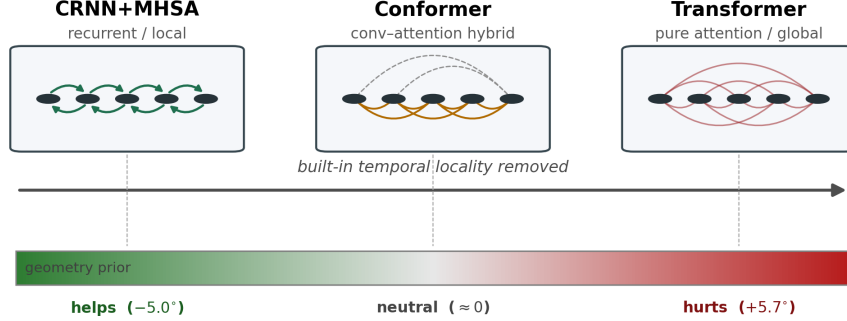


Figure 1: The temporal-locality spectrum and our hypothesis. As the temporal backbone changes from a recurrent CRNN (strong built-in locality) through a convolution–attention Conformer (hybrid) to a pure-attention Transformer (global all-to-all mixing), the value of the hand-crafted array-geometry prior is predicted—and measured—to move monotonically from helpful, through near-neutral, to harmful for directional accuracy. Numbers are the in-domain extreme effects on DOAE_{CD} (Sec. 4.2).

mension 512). It restores locality *within* an attention model, and so is expected to behave between the two extremes.

- **Transformer**, the most *global* backbone: a linear projection followed by a four-layer pre-norm Transformer encoder [14] with eight heads, which removes recurrence and convolution-in-time entirely and mixes all frames and channels globally through self-attention.

All three consume the identical 50-frame convolutional sequence and feed the identical Multi-ACCDDOA heads, so the comparison isolates temporal mixing. The ordering $\text{CRNN} \rightarrow \text{Conformer} \rightarrow \text{Transformer}$ is thus a monotonic decrease in built-in locality, which is the axis along which we predict the prior’s effect to vary. Crossing the three architectures with the two modalities gives six cells; within each, the geometry toggle and its controls yield

Table 1: The grid and task IDs. “full”/“no_geom” is the geometry-bias toggle; controls remove channel attention. All cells use five seeds.

Architecture	Modality	full	no_geom	control
CRNN+MHSA	MIC	110	111	112 / 113
CRNN+MHSA	FOA	130	131	100
Conformer	MIC	161	162	160
Conformer	FOA	171	172	170
Transformer	MIC	141	142	140
Transformer	FOA	151	152	150

the task identifiers in Table 1. Every cell is run with five random seeds.

3.4. Metrics, statistical protocol, and probing

Metrics. We report the official DCASE 2024 metrics: location-aware F_{20° (detection F -score counting a detection correct only within 20°), class-dependent localization error DOAE_{CD} (degrees), location recall, relative distance error (RDE), and the aggregate SELD score. Our headline metric is DOAE_{CD} —the directional accuracy that is the core figure of merit for any direction-of-arrival estimator and exactly what an array-geometry prior is meant to improve; RDE serves as the orthogonal distance axis used to test whether the architecture that modulates direction also modulates distance.

Paired contrasts. Within each cell we match the `full` and `no_geom` runs by seed and summarize the geometry effect as the paired difference $\Delta = \text{full} - \text{no_geom}$, reported with a paired t -test, a Wilcoxon signed-rank test, the standardized effect size Cohen’s d_z , and a bias-corrected bootstrap 95% confidence interval (5000 resamples).

Primary test. Because individual per-cell contrasts have limited power at $n = 5$, our hypothesis does not rest on any single t -test, but accounts for variance explicitly rather than over-interpreting a few runs [21]. Our *primary* test is a factorial Type-II ANOVA over the per-seed metrics that asks whether the geometry effect *interacts* with the temporal architecture. The focal test contrasts the two locality extremes (recurrent vs. pure-attention) in a modality $\times$ architecture $\times$ prior model. Because the same five seeds are matched across cells, we corroborate this interaction with a within-seed paired second-difference test (and confirm with a per-seed random-intercept mixed model that no extra between-seed variance is present), and we test the full graded hypothesis with an ordered Jonckheere–Terpstra trend test across all three backbones rather than only the extremes. We also fit the unordered three-level factorial including the Conformer, and report all of these. We do not correct across the secondary per-cell contrasts, which we treat as descriptive; the confirmatory claim is the pre-specified architecture $\times$ prior interaction and its monotone trend.

Power and the $n = 5$ budget. Five seeds is a deliberate compute trade-off—the grid spans six cells plus no-GCA controls, each retrained and evaluated on two external corpora—and the paired design recovers most of the power a small n would cost, because the large between-seed variance cancels in the within-seed difference Δ . Our confirmatory statistics are therefore defined at the *grid* level: the focal interaction uses all 40 per-seed observations of the recurrent/Transformer sub-grid and the trend test all 60, targeting the large effects we observe ($|d_z| \approx 0.7$ – 1.0 at the extremes, for which a paired $n = 5$ test retains usable power) rather than small per-cell differences.

We treat the independent-dataset replication (Section 4.4) as the decisive evidence rather than any single in-domain p -value.

External validity. Every effect is re-examined out of distribution. We evaluate the STARSS23-trained checkpoints zero-shot on the independent STARSS22 recordings [22] (distance-blind, as STARSS22 lacks distance labels) and stress-test them on the synthetic TAU-NIGENS-SSE-2021 corpus [23], mapping the overlapping classes to the STARSS taxonomy. An effect that reproduces on an independent recording campaign is far stronger evidence than any in-domain p -value.

Mechanism probe. To ask *why* a cell behaves as it does, we linearly probe the representation entering the temporal model. Post-convolutional features are pooled over frequency, and a ridge regressor predicts (sin, cos) of azimuth and elevation on single-source frames using five-fold cross-validation split by recording. The resulting angular MAE measures how *linearly decodable* direction is in the shared representation; comparing **full** against **no_geom** isolates the prior’s effect on linear representation content, independent of the decoder, and tests whether any behavioural effect is mirrored by a change in linearly accessible spatial information.

4. Experiments

4.1. Datasets and reproduction

We train and evaluate in-domain on **STARSS23** (DCASE 2024 Task 3 development set)—a corpus of real, recorded reverberant acoustic scenes in furnished rooms—which provides both MIC and FOA formats with 13 sound classes. For external validity we evaluate zero-shot on **STARSS22**

(DCASE 2022 Task 3 development-test, 54 clips; distance-blind matching, as STARSS22 lacks distance labels) and stress-test on the synthetic **TAU-NIGENS-SSE-2021** corpus, remapping the seven classes that overlap with the STARSS taxonomy. Before ablating we confirm pipeline parity: the FOA reproduction (five seeds) gives $F_{20^\circ} = 13.21 \pm 0.63\%$ against the published 13.10%, with $\text{DOAE}_{\text{CD}} = 41.5 \pm 5.5^\circ$ and $\text{RDE} = 0.27 \pm 0.01$, inside challenge variance. Subsequent differences are thus attributable to the toggled factors, not a mis-implemented baseline.

4.2. Main result: the effect is graded by inductive bias

Table 2 reports the headline contrast, $\Delta\text{DOAE}_{\text{CD}} = \text{full} - \text{no_geom}$, seed-matched within each cell ($n = 5$) and grouped by temporal backbone; Fig. 2 visualizes it. The effect on direction follows a **backbone-ordered pattern**: the recurrent cells are non-positive overall, the Transformer cells are positive overall, and the convolution-attention Conformer occupies a near-neutral middle position. The two extremes remain the load-bearing evidence in this ordering: the prior improves localization by 5.0° in FOA+CRNN ($d_z = -1.03$) and degrades it by 5.7° in MIC+Transformer ($d_z = +0.71$). The Conformer does not behave like a second Transformer. Instead, its paired summaries place it between the two extremes: mildly helpful on MIC and near-neutral on FOA.

At $n = 5$ the individual per-cell contrasts are trends rather than significant in isolation (e.g. FOA+CRNN $p = 0.082$, MIC+Transformer $p = 0.186$). *These single-cell p-values are not our primary evidence*; the claim is about the *ordering* across the grid, which we test jointly with a factorial interaction (Section 4.3) and corroborate on an independent dataset where the

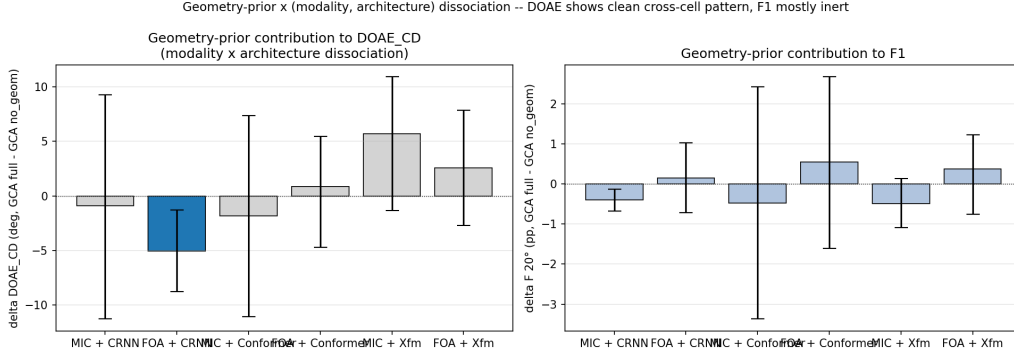


Figure 2: The geometry-prior effect on DOAE_{CD} across the modality $\times$ backbone grid. The sign of the effect tracks temporal inductive bias: it helps (negative) under recurrence, is near-neutral for the Conformer, and hurts (positive) under the Transformer, the extremes being FOA+CRNN and MIC+Transformer.

recurrent benefit reaches full significance (Section 4.4). This is deliberately a *direction-localization* claim: the same archived per-seed table does not show a corresponding architecture-ordered pattern on F_{20° or aggregate SELD score, which we therefore treat as secondary diagnostics rather than as the endpoint defining the prior’s sign.

One honest caveat about the middle of the ordering is that the Conformer is also the weakest detector in our grid (location-aware F_{20° of 4–7% and the worst SELD scores, vs. 9–13% for the extremes). Its small geometry effect may therefore reflect both intermediate locality and a weaker operating point. We accordingly treat the Conformer as a *consistency check on the ordering*, not as the load-bearing evidence for it: the robust contrast is still recurrent-versus-Transformer, while the Conformer matters because it stays in the middle rather than collapsing to the Transformer-like harmful regime.

Table 2: Main result, grouped by backbone: paired geometry-prior contrast on class-dependent localization error, $\Delta\text{DOAE}_{\text{CD}} = \text{full} - \text{no_geom}$ (degrees; negative = prior helps), $n = 5$ each. The per-backbone mean sign ordering $\text{CRNN} \leq \text{Conformer} \leq \text{Transformer}$ holds after averaging over modality; the recurrent and Transformer extremes also order as predicted within each modality. Source: `path_c_2x2`.

Backbone	Modality	ΔDOAE	t (p)	d_z
<i>CRNN+MHSA (recurrent): prior helps</i>				
CRNN	MIC	−0.88	−0.15 (0.887)	−0.07
CRNN	FOA	−5.02	−2.31 (0.082)	−1.03
<i>Conformer (conv-attention hybrid): intermediate</i>				
Conformer	MIC	−1.83	−0.35 (0.745)	−0.16
Conformer	FOA	−0.57	−0.26 (0.811)	−0.11
<i>Transformer-only: prior hurts</i>				
Transformer	MIC	+5.70	+1.60 (0.186)	+0.71
Transformer	FOA	+2.57	+0.86 (0.437)	+0.39

4.3. Primary test: *architecture*×*prior* interaction

Our primary hypothesis test asks whether the sign ordering of Table 2 is a genuine interaction rather than a collection of independent per-cell coincidences. We report it three complementary ways on the per-seed DOAE_{CD} , all pointing the same direction (Source: `path_c_trend_mixed`).

(i) *Factorial interaction, seed-matched.* A Type-II factorial ANOVA restricted to the recurrent and Transformer extremes finds that the architecture by prior interaction is significant ($F = 4.62$, $p = 0.039$; $n_{\text{obs}} = 40$): the prior’s effect on direction depends on whether the backbone is recurrent or pure-

attention. Because the same five seeds are matched across cells, we re-express this interaction as a within-seed paired *second difference*, $D = \Delta_{\text{CRNN}} - \Delta_{\text{Xfm}}$ per matched (modality, seed), where $\Delta = \text{full} - \text{no_geom}$; D is reliably negative ($\bar{D} = -7.1^\circ$, $t = -2.59$, $p = 0.029$, $d_z = -0.82$), i.e. the prior helps the recurrent backbone about 7° more than it helps the Transformer. A linear mixed model with a per-seed random intercept estimates that variance at the zero boundary ($\text{ICC} \approx 0$ for DOAE_{CD}), so the design carries no extra between-seed structure and the fixed-effects estimate is the appropriate one.

(ii) *Monotone trend across all three backbones.* Rather than discard the intermediate Conformer, we test the *ordered* hypothesis that the geometry effect Δ increases monotonically along the locality axis from CRNN to Conformer to Transformer. A Jonckheere–Terpstra trend test on the per-seed Δ (pooled over modality) is a one-sided trend at $z = +1.60$, $p = 0.055$, with a parametric slope of $+3.5^\circ$ per locality step ($p = 0.073$); the trend is clearest in Ambisonics (one-sided $p = 0.051$) and is the weakest in the microphone array, where the Conformer is itself mildly helpful. The monotone ordering of the per-backbone means ($-2.9^\circ \rightarrow -0.5^\circ \rightarrow +4.1^\circ$) is therefore a genuine—if, at $n = 5$, marginal—trend, not an artifact of selecting the two extremes.

(iii) *Unordered omnibus.* When the Conformer is added as a third level in the *unordered* factorial ($n_{\text{obs}} = 60$), the three-level architecture $\times$ prior term is not significant ($F = 1.94$, $p = 0.15$), exactly as expected when a near-zero middle category is inserted between two opposing extremes: an unordered omnibus spends power on a quadratic contrast the data do not exhibit, whereas the ordered trend test in (ii) retains it. The prior main effect is null throughout ($p > 0.6$), confirming the effect is conditional rather

than a universal benefit or harm. (On the detection metric F_{20° the design shows strong modality $F = 36.9$ and architecture $F = 79.2$ main effects, consistent with FOA and recurrence detecting events more readily overall; these concern detectability, not the prior.)

4.4. *Anchor: the ordering replicates zero-shot on STARSS22*

The grid ordering is not a STARSS23 artifact. Evaluating the STARSS23-trained checkpoints zero-shot on STARSS22 (Table 3 and Fig. 3, $n = 5$) reproduces the same architecture ordering out-of-domain: both recurrent cells help, the Conformer remains near-neutral, and both Transformer cells hurt. The forest plot makes the agreement visible—in-domain and zero-shot estimates share the same sign in every cell, and the effect marches monotonically from the “helps” to the “hurts” region as locality is removed. The single *significant* cross-dataset effect is the recurrent benefit on Ambisonics, FOA+CRNN $\Delta\text{DOAE}_{\text{CD}} = -4.76^\circ$ ($d_z = -1.50$, $p = 0.029$), which matches its in-domain sign and magnitude (-5.02°)—a benefit reproducing on an independent recording campaign is strong evidence that it is a property of the learned model, not of one test set. The MIC+Transformer harm reproduces directionally ($+5.58^\circ$, $d_z = +0.77$) but, like in-domain, does not reach significance on its own at $n = 5$. The cross-dataset per-backbone means trace the same monotonic path as in-domain (helpful $\rightarrow$ near-neutral $\rightarrow$ harmful).

4.5. *Second injection check: the ordering is bounded by mechanism*

The GCA experiments above isolate a geometry term inside channel-attention keys. To test whether the architecture ordering is merely an artifact of that injection site, we also implement an orthogonal geometry path,

Table 3: Anchor. STARSS22 zero-shot, geometry-prior contrast on DOAE_{CD} (degrees; negative = helps), $n = 5$. The in-domain ordering holds out-of-domain; the FOA+CRNN benefit is significant. Source: `path_c_cross_starss22`.

Backbone	Modality	Δ DOAE (cross)	in-domain
CRNN	MIC	-4.77 ($d_z = -0.42$)	-0.88
CRNN	FOA	-4.76 ($d_z = -1.50$, $p=.029$)	-5.02
Conformer	MIC	-3.20 ($d_z = -0.48$)	-1.83
Conformer	FOA	-0.42 ($d_z = -0.06$)	-0.57
Transformer	MIC	$+5.58$ ($d_z = +0.77$)	$+5.70$
Transformer	FOA	$+1.72$ ($d_z = +0.39$)	$+2.57$

`convbias`: a fixed geometry descriptor is projected with a learned linear layer into one scalar per front-end convolutional filter and added as an early feature-map bias. The `full` and `no_geom` variants keep identical parameter counts; the latter runs the same projection but with the geometry input zeroed. This makes the contrast a matched-capacity test of geometry content rather than extra parameters.

Table 4 shows that `convbias` should be interpreted as a sensitivity check rather than a confirmatory cross-mechanism replication. On FOA Conformer, the first matched runs remain close to zero, which is broadly consistent with the GCA middle-point reading, but the measured contrast moves under stricter rerun and checkpoint-selection paths. FOA+Transformer is more unstable still, changing sign across reruns, and MIC+Transformer reverses relative to GCA. We therefore use `convbias` to bound the claim: the

Table 4: Second injection check on DOAE_{CD}. GCA injects geometry inside channel-attention keys ($n = 5$). `convbias` injects geometry as an early per-filter convolutional feature bias ($n = 3$ per run). Each entry is $\Delta = \text{full} - \text{no_geom}$ in degrees (negative = helps). Rather than confirming a mechanism-invariant effect, the `convbias` reruns act as a sensitivity audit: the Conformer stays near-neutral in early runs, whereas Transformer contrasts and deterministic SELD-selection reruns reveal substantial mechanism and checkpoint sensitivity.

Modality	Backbone	GCA	run 1	GPU rerun	det. SELD	Reading
FOA	CRNN	-5.02	-1.60	—	—	weak help
FOA	Conformer	-0.57	+0.21	+0.31	+7.59	checkpoint-sensitive
FOA	Transformer	+2.57	+9.55	-2.89	+4.81	unstable
MIC	Transformer	+5.70	-3.79	—	—	sign reversal

primary architecture-ordered result is stable for the controlled GCA intervention, whereas this second injection site exposes training-selection and mechanism sensitivity that is not yet stable enough to support a stronger mechanism-invariant statement.

4.6. Mechanism probe: the effect is behavioural, not a linear feature shift

A natural hypothesis is that the prior changes how linearly direction is encoded in the shared post-convolutional representation. We test this by linearly probing those features (frequency-pooled, ridge regression to $(\sin, \cos)$ of azimuth and elevation on single-source frames, file-wise five-fold CV) and comparing `full` against `no_geom` (Table 5). At $n = 5$, **no cell shows a significant change** in linearly decodable direction ($|d_z| \leq 0.53$, all $p > 0.2$); the largest movement is a non-significant increase in MIC+Conformer. We report this as an informative negative result: the prior’s behavioural effect

on SELD output is *not* mirrored by a measurable shift in the linear direction content of the post-conv features, so the mechanism does not reduce to “the prior adds/removes decodable direction information.” This null is non-trivial. The geometry bias gates the input channels *upstream* of the shared convolutional front-end, and the `full` and `no_geom` variants train their own convolutional weights, so the probed post-convolutional features could in principle diverge; that they do not differ in linearly decodable direction means the prior leaves the linearly accessible spatial content of the shared representation essentially unchanged. Its behavioural effect must therefore reside in nonlinear or distributed feature changes the linear probe cannot capture, or in how each temporal stack *uses* that representation—the latter consistent with the architecture-dependence we observe. (We caution that an earlier three-seed probe suggested a Transformer-specific loss; it did not survive the move to five seeds, a useful reminder of the fragility of small- n representational claims.)

4.7. Boundary conditions and supporting ablations

Distance follows a different rule. Beyond direction, the prior’s effect on the third (distance) dimension does *not* track the temporal architecture (Table 6): relative distance error improves by effect size in *all three* Ambisonic cells—FOA+CRNN $\Delta\text{RDE} = -0.024$ ($d_z = -1.43$, $p = 0.033$), FOA+Transformer -0.020 ($d_z = -1.00$), FOA+Conformer -0.010 ($d_z = -0.35$)—and is null in all three microphone-array cells ($|d_z| \leq 0.26$, all $p > 0.5$), with no architecture $\times$ prior interaction ($F = 0.20$, $p = 0.82$; paired CRNN–Xfm second difference $p = 0.52$). The architecture that governs the *direction* effect is thus inert on distance, where the benefit

Table 5: Mechanism probe. Linear-probe angular MAE (degrees; lower = direction more linearly decodable), Δ = full – no_geom, $n = 5$. No cell shows a significant geometry effect. Source: `path_c_probe`.

Backbone	Modality	full	no_geom	Δ
CRNN	MIC	28.50	28.37	+0.12
CRNN	FOA	20.62	20.89	−0.27
Conformer	MIC	29.88	28.35	+1.53
Conformer	FOA	18.69	19.28	−0.59
Transformer	MIC	26.11	26.27	−0.16
Transformer	FOA	17.93	17.89	+0.05

is instead concentrated in Ambisonics. We are careful not to overclaim the second half of this contrast: at $n = 5$ the modality×prior interaction on distance is itself only a non-significant trend ($F = 0.39$, $p = 0.54$; paired FOA–MIC second difference $p = 0.54$), so the distance pattern is an effect-size dissociation—architecture selectively modulates direction and not distance—rather than a second statistically confirmed interaction. Source: `path_c_dissociation_test`.

Boundary condition (synthetic transfer). On the synthetic TAU-NIGENS corpus (seven mapped classes, zero-shot; CRNN and Transformer cells, $n = 3$ –5) detection collapses to near-floor for every cell (kept-class $F \approx 1$ –2%, $\text{DOAE}_{\text{CD}} \approx 60^\circ$) under the heavy synthetic domain shift, and at that floor the geometry-prior contrasts are uniformly null. Notably the in-domain MIC+Transformer harm does *not* transfer: the synthetic $\Delta\text{DOAE}_{\text{CD}}$ is $+1.1^\circ$ ($d_z = +0.17$) versus $+5.7^\circ$ in-domain, and the MIC+CRNN and

Table 6: Distance is modality-patterned, not architecture-conditional. Geometry-prior contrast on relative distance error, $\Delta\text{RDE} = \text{full} - \text{no_geom}$ ($n = 5$; negative = helps). All Ambisonic cells improve by effect size; all microphone-array cells are null—orthogonal to the architecture-conditional direction effect of Table 2, though the modality $\times$ prior interaction is itself only a non-significant trend at $n = 5$ (see text). Source: `path_c_2x2`.

Backbone	Modality	ΔRDE	d_z
CRNN	MIC	−0.016	−0.26
CRNN	FOA	−0.024 ($p=.033$)	−1.43
Conformer	MIC	−0.008	−0.17
Conformer	FOA	−0.010	−0.35
Transformer	MIC	−0.002	−0.04
Transformer	FOA	−0.020	−1.00

FOA contrasts are likewise null ($|d_z| \leq 0.6$). We read this as a boundary condition: a domain shift severe enough to erase detection also erases the subtle prior effect, whereas the real-data architecture ordering (Sections 4.2–4.4) does not depend on it.

Supporting controls. A per-class breakdown (13 classes, all cells) shows detectable F -score concentrating in speech, music, and domestic-sound classes, so the architecture ordering is a property of the localization representation, not of a few classes’ detectability. A vanilla squeeze-and-excitation control in MIC+CRNN confirms that the *geometric* bias, not channel attention itself, drives the contrast.

5. Discussion

5.1. Inductive bias grades the sign

The temporal backbone’s inductive bias, not the input modality alone, determines *whether* the prior helps direction—and it does so as a *gradient*, not a switch. A recurrent model has limited capacity to recombine channels over time, so the geometry bias acts as a useful scaffold; a Transformer already mixes all channels globally, so the same bias acts as a redundant or mismatched constraint; the convolution–attention Conformer sits between them and correspondingly shows a near-neutral to weakly helpful effect. The significant recurrent-versus-Transformer architecture $\times$ prior interaction (Section 4.3) and the ordered grid-level pattern are both explained by this single locality axis, with modality mainly modulating effect size.

5.2. Why the recurrent benefit is the anchor

The single most robust effect is the recurrent *benefit* on Ambisonics. The geometry bias of Eq. (1) aligns the channel attention with the directional structure a low-capacity recurrent model would otherwise have to learn from scarce real data, yielding the 5.0° in-domain FOA+CRNN gain (a trend, $p = 0.082$) that replicates at full significance zero-shot on STARSS22 ($d_z = -1.50$, $p = 0.029$; Table 3). Consistency of sign and magnitude on an independent recording campaign indicates a property of the learned model, not of one test split. Symmetrically, the Transformer is harmed—forcing a global-mixing model’s channel attention toward a fixed, geometry-derived pattern mismatches the channel statistics of real reverberant rooms—and this harm reproduces directionally in-domain (+5.7°) and cross-dataset (+5.6°),

though at $n = 5$ it does not reach significance as a standalone contrast. The Conformer matters because it does *not* collapse to this harmful regime, yet it also does not become a strong positive case; it is better interpreted as a near-neutral middle point.

5.3. *The mechanism is behavioural, not a feature shift*

We initially hypothesised that the prior changes the *linear* direction content of the shared representation. The $n = 5$ probe rejects this: no cell shows a significant change in linearly decodable direction (Table 5). This null is not a foregone conclusion—the geometry bias gates the input channels *upstream* of the convolutional front-end and each variant trains its own convolutional weights, so the post-convolutional features could in principle diverge. That they remain near-identical means the prior leaves the linearly accessible spatial content essentially unchanged; its behavioural effect must instead reside in nonlinear or distributed changes the probe cannot see, or in how each temporal stack *uses* those features—consistent with the locality account above. This also corrects a fragile three-seed probe from an earlier draft that did not survive added seeds.

5.4. *The cross-dataset picture*

The full architecture ordering survives domain shift to STARSS22 (Table 3): recurrent cells help, the Conformer stays near-neutral, and Transformer cells hurt. MIC+CRNN, null in-domain, is mildly *helped* cross-dataset (-4.77°), suggesting the bias can also act as a weak regularizer toward array-consistent solutions under shift—reinforcing the graded account: the prior’s value depends jointly on backbone and deployment regime.

5.5. *The injection mechanism bounds the claim*

The `convbias` check sharpens the interpretation by limiting it. Moving the prior from channel-attention keys to an early convolutional feature bias does not deliver a clean mechanism-invariant replication. Instead, the FOA Conformer stays near-neutral in early runs but shifts under stricter rerun and checkpoint-selection paths; FOA Transformer changes sign across reruns; and the MIC+Transformer contrast reverses relative to GCA. This rules out stronger claims such as “geometry priors always hurt Transformers” or “the Conformer is mechanism-invariantly neutral.” The design lesson is narrower and more useful: both the temporal backbone and the injection site must be ablated, because a hand-crafted prior can change role when imposed before convolution rather than inside channel attention, and checkpoint selection can materially move the measured directional error.

5.6. *Practical guidance*

Do not add an array-geometry prior by default. In this study it is most useful for a lower-capacity recurrent model, near-neutral in a Conformer, and capable of harming a pure-attention Transformer; moreover, alternative injection paths can themselves be unstable. In practice, the right default is not “use the prior” or “avoid the prior” but “ablate the prior with a matched-capacity, seed-matched test before shipping.”

5.7. *Limitations*

(i) *Absolute performance is low* (F_{20° of 9–13%, DOAE_{CD} of 36–50°): we deliberately study a compact official baseline on a single 4 GB GPU, not a leaderboard system. Because the one-bit intervention holds capacity, data,

and recipe fixed, the *relative* effects remain interpretable; we do not claim the magnitudes transfer unchanged to 50M-parameter ensembles. *(ii) Modest seed counts* ($n = 5$ per cell) keep most per-cell contrasts at the level of trends; our conclusions rest on the grid-level ANOVA interaction, its trend, and the significant cross-dataset replication, not on any single in-domain t -test, and larger budgets would sharpen the per-cell estimates. *(iii) Synthetic transfer is a floor*: under the severe TAU-NIGENS domain shift detection collapses for all cells, so that corpus bounds rather than extends the effect; the load-bearing external validity is the STARSS22 replication, which covers both modalities. *(iv) Backbone, mechanism, and determinism*: we study three backbones spanning the locality spectrum but a single front-end and recipe, with one primary geometry mechanism and one smaller second-injection check. That check is sensitive to reruns and checkpoint selection, so it bounds rather than confirms the primary GCA conclusion. A deterministic training stack and larger seed budget would be needed to make stronger claims about alternative injection sites.

6. Conclusion

We asked when a hand-crafted array-geometry prior helps SELD, using a controlled one-bit ablation across modality and three temporal backbones. The effect on direction is **graded by temporal inductive bias**: recurrent cells shift non-positively, Transformer cells positively, and the Conformer occupies a near-neutral middle position rather than a second harmful extreme. The most robust effect is the FOA recurrent benefit, which replicates zero-shot on STARSS22 at full significance; the strongest harm is the MIC

Transformer contrast. A factorial ANOVA confirms the recurrent-versus-Transformer architecture $\times$ prior interaction, and a controlled probe shows that the effect is behavioural rather than a shift in linearly decodable direction. Orthogonally, the architecture that governs direction is inert on distance—a dissociation, not two crossed interactions. A second injection mechanism, `convbias`, does not strengthen this claim into a mechanism-invariant law; instead it exposes rerun, checkpoint-selection, and injection-site sensitivity, thereby bounding the primary GCA result. The practical conclusion is therefore conditional rather than universal: explicit geometry priors are useful design choices only in the context of a particular backbone and injection path, and should be ablated rather than assumed. We release a reproducible multi-seed pipeline so future work can audit other priors, injection sites, and backbones with the same protocol.

Appendix A. Complete per-cell results

Table A.7 reports every trained configuration in the grid. It covers the six modality $\times$ backbone cells in their `full`, `no_geom`, and `no-GCA` variants, plus the FOA baseline and the squeeze-and-excitation control, with all four primary metrics averaged over five seeds (mean $\pm$ std). These are the raw numbers underlying the paired contrasts in the main text. The FOA no-GCA baseline (row 100) matches the published DCASE 2024 figure (13.1%), confirming pipeline parity.

Appendix B. Per-class detection

Table B.8 reports class-wise location-aware F_{20° for the six geometry-aware (**full**) cells. Detectable performance concentrates in a handful of well-represented classes—speech (male/female), music, domestic sounds, and water tap—while sparse or transient classes (clapping, door, knock) sit at the floor across *all* cells. The architecture ordering of the geometry effect is therefore a property of the shared localization representation rather than of any single class’s detectability.

CRedit authorship contribution statement

Author contributions withheld for double-blind review; the CRediT taxonomy statement will be completed in the camera-ready version.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

This study uses the publicly available STARSS23 and STARSS22 datasets and the NIGENS sound-event corpus. The complete training and evaluation pipeline, together with the trained checkpoints, will be released publicly upon publication.

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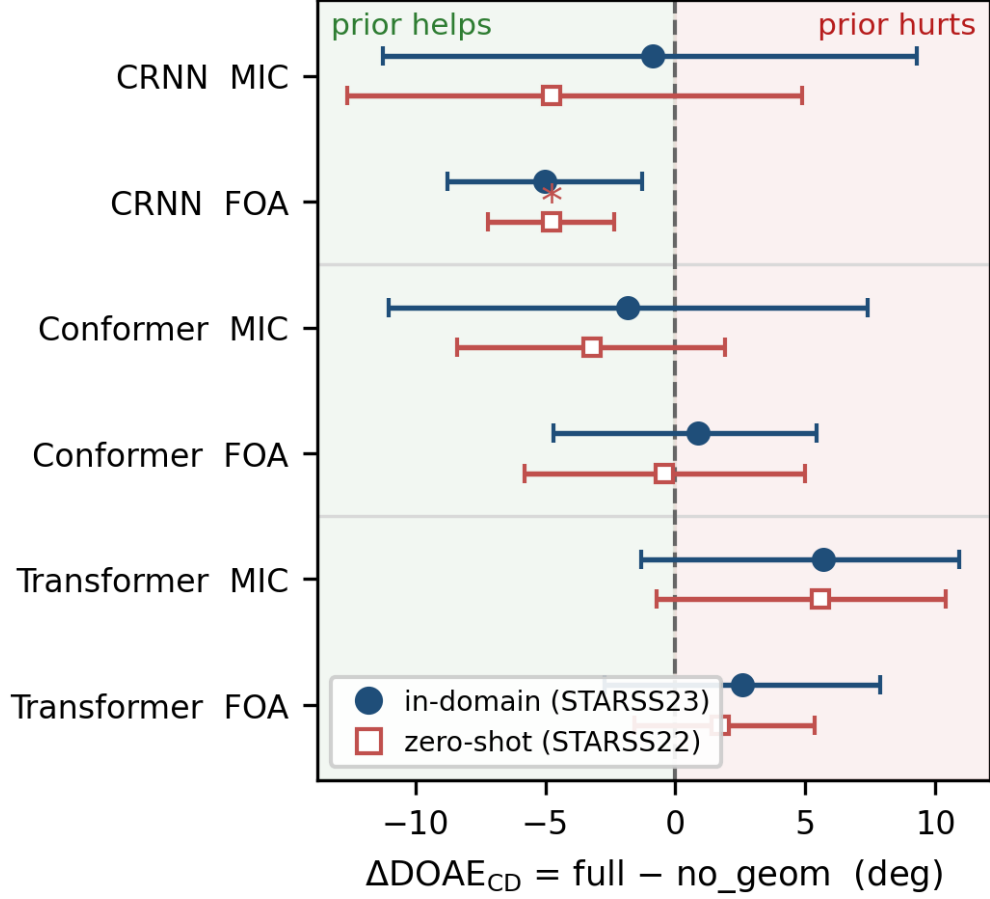


Figure 3: Geometry-prior effect on directional error ($\Delta\text{DOAE}_{\text{CD}} = \text{full} - \text{no_geom}$; negative = helps) per cell, ordered by temporal locality, with bootstrap 95% CIs. Filled circles are in-domain (STARSS23); open squares are zero-shot on STARSS22; “*” marks $p < 0.05$. The effect shifts monotonically from the helps region (recurrent) to the hurts region (Transformer), and the in-domain and zero-shot estimates agree in sign for every cell—the recurrent Ambisonic benefit being significant out-of-domain.

Table A.7: Complete in-domain (STARSS23 dev-test) results for all configurations, mean $\pm$ std over five seeds. F_{20° and SELD score: higher/lower is better respectively; DOAE_{CD} and RDE: lower is better. “full” = geometry-aware GCA, “no_geom” = GCA with the geometry bias disabled, “no-GCA” = channel attention removed, “SE” = vanilla squeeze-and-excitation.

ID	Configuration	n	F_{20° (%)	DOAE _{CD} ($^\circ$)	RDE	SELD
100	FOA no-GCA (baseline)	5	13.21 ± 0.63	41.5 ± 5.5	0.27 ± 0.01	0.526 ± 0.009
110	MIC CRNN full	5	9.59 ± 0.41	45.4 ± 6.4	0.28 ± 0.05	0.547 ± 0.026
111	MIC CRNN no_geom	5	9.99 ± 0.54	46.3 ± 7.5	0.30 ± 0.03	0.545 ± 0.024
112	MIC CRNN no-GCA	5	9.72 ± 0.46	46.3 ± 4.2	0.29 ± 0.03	0.579 ± 0.031
113	MIC CRNN SE	5	10.06 ± 0.59	44.8 ± 7.3	0.30 ± 0.04	0.558 ± 0.037
130	FOA CRNN full	5	12.79 ± 0.65	40.1 ± 2.2	0.26 ± 0.01	0.531 ± 0.002
131	FOA CRNN no_geom	5	12.64 ± 0.60	45.1 ± 5.9	0.29 ± 0.01	0.538 ± 0.011
160	MIC Conformer no-GCA	5	5.50 ± 3.46	52.2 ± 4.3	0.31 ± 0.04	0.718 ± 0.121
161	MIC Conformer full	5	3.59 ± 2.35	47.1 ± 3.9	0.27 ± 0.02	0.687 ± 0.067
162	MIC Conformer no_geom	5	2.56 ± 1.26	48.9 ± 5.5	0.30 ± 0.03	0.713 ± 0.074
170	FOA Conformer no-GCA	5	7.28 ± 3.11	48.6 ± 7.9	0.28 ± 0.05	0.655 ± 0.111
171	FOA Conformer full	5	5.86 ± 0.97	51.2 ± 3.0	0.28 ± 0.02	0.649 ± 0.072
172	FOA Conformer no_geom	5	4.37 ± 2.76	51.8 ± 4.6	0.29 ± 0.03	0.700 ± 0.078
140	MIC Transformer no-GCA	5	9.61 ± 0.44	41.1 ± 5.1	0.29 ± 0.03	0.603 ± 0.032
141	MIC Transformer full	5	9.13 ± 0.75	48.0 ± 5.2	0.29 ± 0.04	0.630 ± 0.055
142	MIC Transformer no_geom	5	9.63 ± 0.59	42.3 ± 3.1	0.30 ± 0.02	0.599 ± 0.030
150	FOA Transformer no-GCA	5	10.95 ± 0.64	36.0 ± 3.1	0.31 ± 0.04	0.643 ± 0.065
151	FOA Transformer full	5	11.13 ± 0.50	42.7 ± 5.1	0.28 ± 0.03	0.598 ± 0.034
152	FOA Transformer no_geom	5	10.75 ± 1.09	40.1 ± 4.2	0.30 ± 0.02	0.631 ± 0.071

Table B.8: Per-class location-aware F_{20° (% , mean over five seeds) for the geometry-aware (full) cells on STARSS23 dev-test. Classes are sorted by overall detectability.

Class	CRNN		Conformer		Transformer	
	MIC	FOA	MIC	FOA	MIC	FOA
Male speech	36.9	40.1	16.5	29.9	32.5	31.9
Female speech	34.4	37.9	14.1	27.8	32.5	29.9
Domestic sounds	24.6	42.0	3.5	3.5	21.7	22.7
Music	18.4	17.1	10.4	7.8	15.5	12.6
Water tap	3.5	14.4	7.2	18.3	9.4	40.9
Laughter	2.7	3.8	1.4	3.1	0.3	1.1
Footsteps	3.1	3.9	0.8	0.0	1.2	2.3
Instrument	2.3	2.7	1.2	0.2	1.1	0.8
Telephone	0.0	2.9	0.2	2.4	4.7	2.4
Bell	0.5	2.2	0.0	1.2	0.6	0.3
Clapping	0.0	0.0	0.0	0.0	0.0	0.0
Door open	0.0	0.0	0.0	0.0	0.0	0.0
Knock	0.0	0.0	0.0	0.0	0.0	0.0